# NexLearn LMS

## Activity Diagram Document

UE23CS352B — Object Oriented Analysis & Design
Domain: Education | Technology: Spring Boot 3.4 + Next.js 15 + MongoDB



| Team Members | Laharish S (PES2UG23CS298) |
|---|---|
| | Kathit Joshi (PES2UG23CS264) |
| | Kavyansh Jain (PES2UG23CS268) |
| | Kumarchandra Edupuganti (PES2UG23CS292) |



## 1. Document Overview

This document presents the Activity Diagrams for all 10 use cases of the NexLearn Learning Management System. Each activity diagram models the complete workflow of its corresponding use case using swimlanes to distinguish between the Actor (Student, Instructor, or Anonymous User) and the Spring Boot System.

Activity diagrams are created using PlantUML. The diagram images are to be inserted in the placeholder sections below after export from the PlantUML editor.

## 2. Use Case Summary

| UC ID | Use Case Name | Primary Actor | Type | Priority |
|---|---|---|---|---|
| UC-01 | User Registration | Student, Instructor | Major | High |
| UC-02 | User Authentication (Login/Logout) | Student, Instructor | Major | High |
| UC-03 | Course Creation | Instructor | Major | High |
| UC-04 | Course Enrollment | Student | Major | High |
| UC-05 | Content Upload & Management | Instructor | Major | High |
| UC-06 | Learning Dashboard & Progress Tracking | Student | Minor | Medium |
| UC-07 | Student Monitoring by Instructor | Instructor | Minor | Medium |

| UC ID | Use Case Name | Primary Actor | Type | Priority |
|---|---|---|---|---|
| UC-08 | Search Courses | Student, Instructor | Minor | Medium |
| UC-09 | User Profile Management | Student, Instructor | Minor | Low |
| UC-10 | Contact & Support | Student, Instructor, Anonymous | Minor | Low |

# 3. Activity Diagram Specifications

## 1. UC-01 — User Registration

| Use Case Reference | |
|---|---|
| **Use Case ID** | UC-01 |
| **Use Case Name** | User Registration |
| **Primary Actor(s)** | Student, Instructor |
| **Type** | Major |
| **Priority** | High |

**Description:** Depicts the end-to-end flow of a new user creating an account, including input validation, duplicate email check, password hashing, and MongoDB persistence. Swimlanes: Student/Instructor | Spring Boot System.

**Notes:** *The System lane handles all validation and persistence. The user lane only performs UI interactions.*

## [Activity Diagram — UC-01: User Registration]

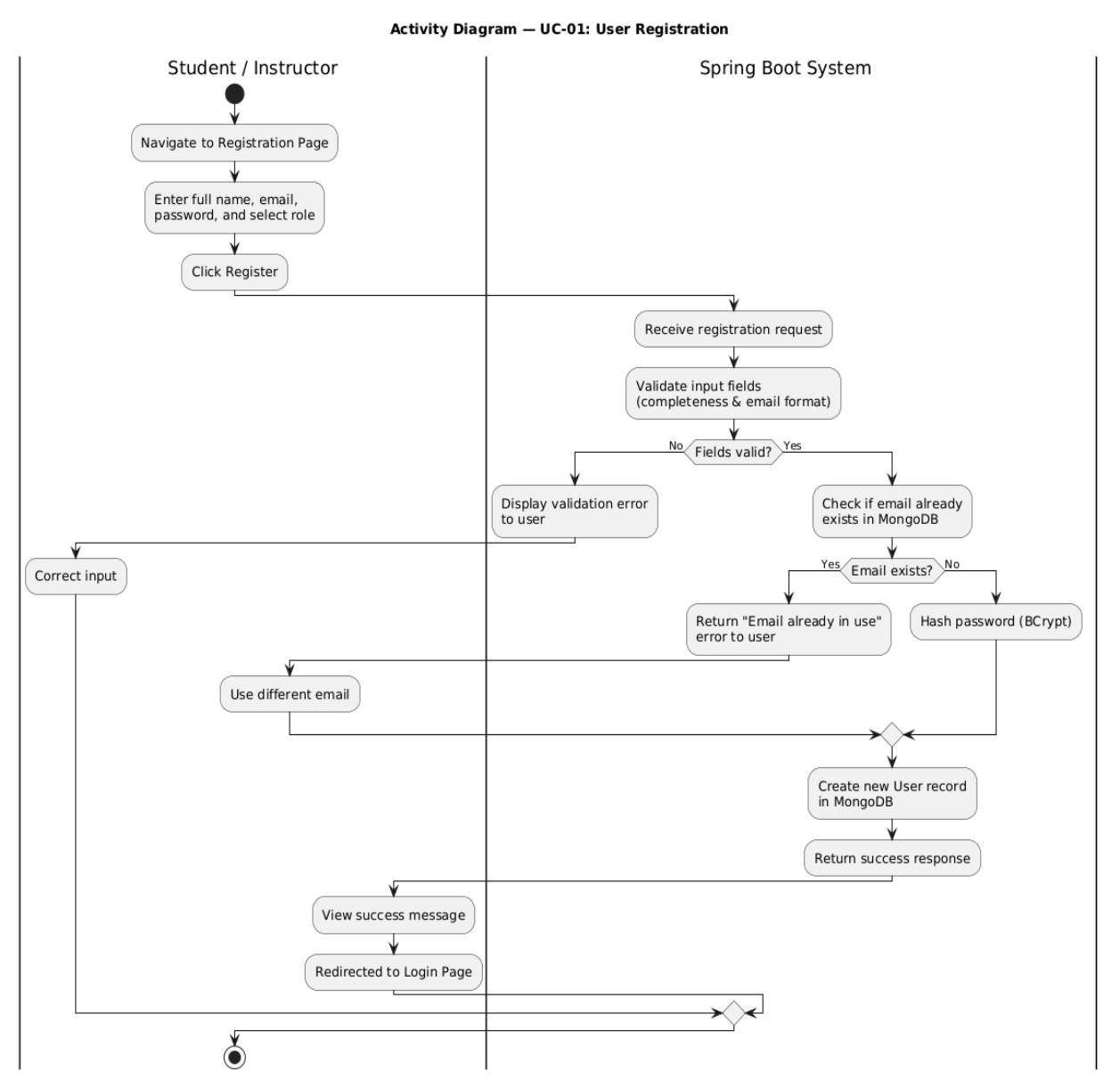


*Figure 1: Activity Diagram — UC-01: User Registration*

## 2. UC-02 — User Authentication (Login/Logout)

| Use Case Reference | |
|---|---|
| **Use Case ID** | UC-02 |
| **Use Case Name** | User Authentication (Login/Logout) |
| **Primary Actor(s)** | Student, Instructor |
| **Type** | Major |
| **Priority** | High |

**Description:** Shows the complete login flow — credential validation, JWT generation — and the logout flow. Swimlanes: Student/Instructor | Spring Boot System.

**Notes:** *JWT is stored client-side. Logout only clears the client token; no server-side session is maintained.*

## [Activity Diagram — UC-02: User Authentication (Login/Logout)]

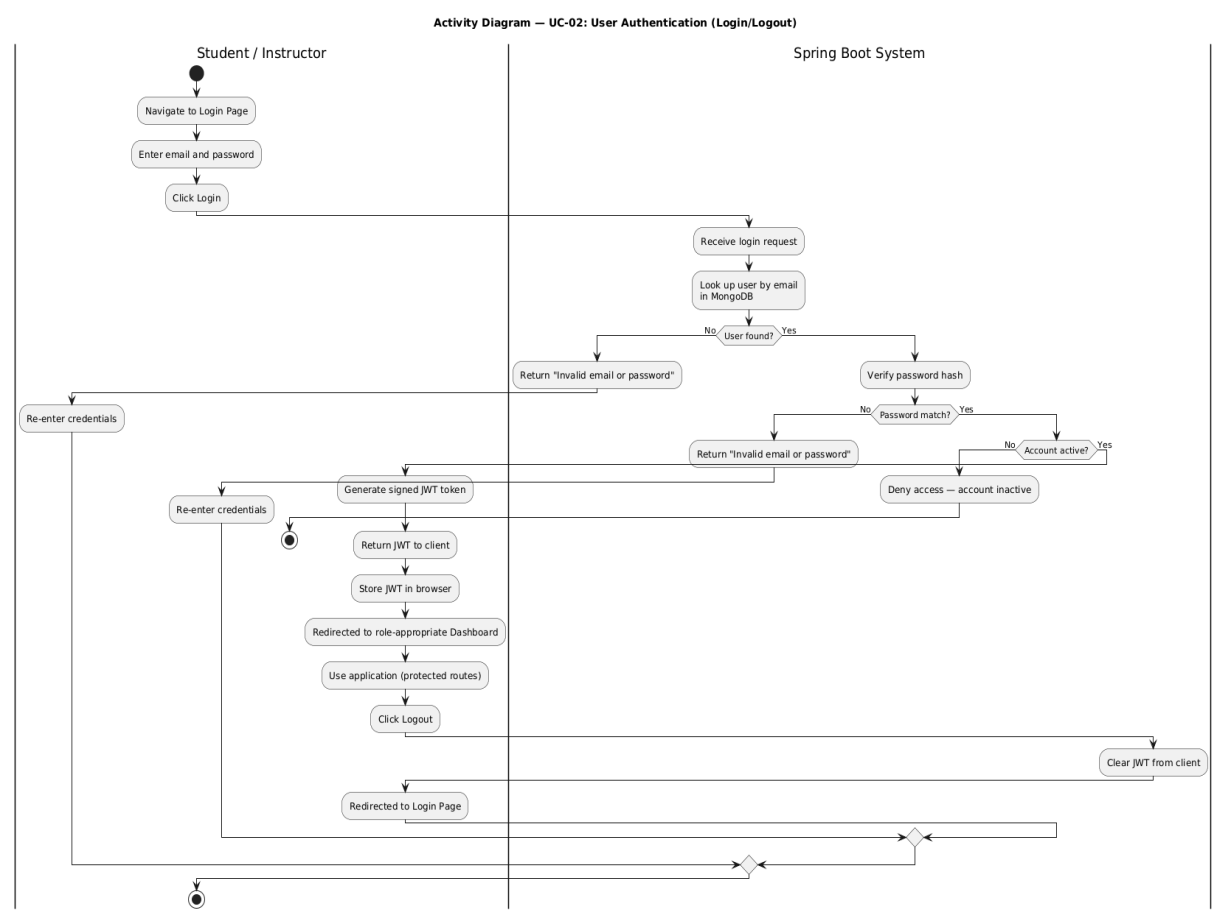


*Figure 2: Activity Diagram — UC-02: User Authentication (Login/Logout)*

## 3. UC-03 — Course Creation

| Use Case Reference | |
|---|---|
| **Use Case ID** | UC-03 |
| **Use Case Name** | Course Creation |
| **Primary Actor(s)** | Instructor |
| **Type** | Major |
| **Priority** | High |

**Description:** Illustrates the instructor creating a course: JWT validation, field validation, MongoDB persistence, and redirection. Swimlanes: Instructor | Spring Boot System.

**Notes:** *Only authenticated instructors can reach the create-course endpoint. JWT is validated before any field processing.*

## [Activity Diagram — UC-03: Course Creation]

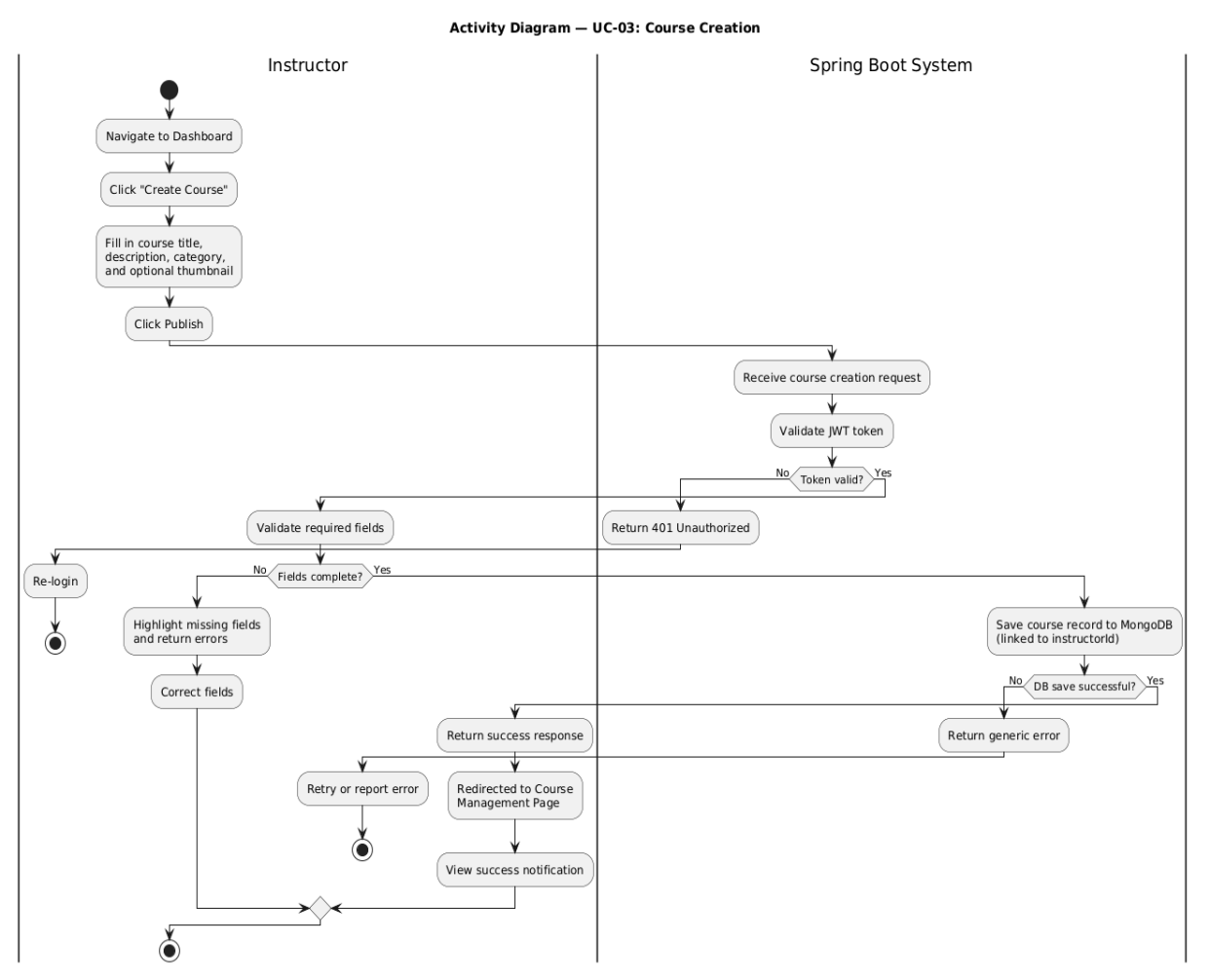


*Figure 3: Activity Diagram — UC-03: Course Creation*

## 4. UC-04 — Course Enrollment

| Use Case Reference | |
|---|---|
| **Use Case ID** | UC-04 |
| **Use Case Name** | Course Enrollment |
| **Primary Actor(s)** | Student |
| **Type** | Major |
| **Priority** | High |

**Description:** Models course browsing, course selection, duplicate enrollment prevention, and enrollment record creation. Swimlanes: Student | Spring Boot System.

**Notes:** *The System checks for a pre-existing Enroll document before saving. Duplicate enrollments are rejected with HTTP 409.*

## [Activity Diagram — UC-04: Course Enrollment]

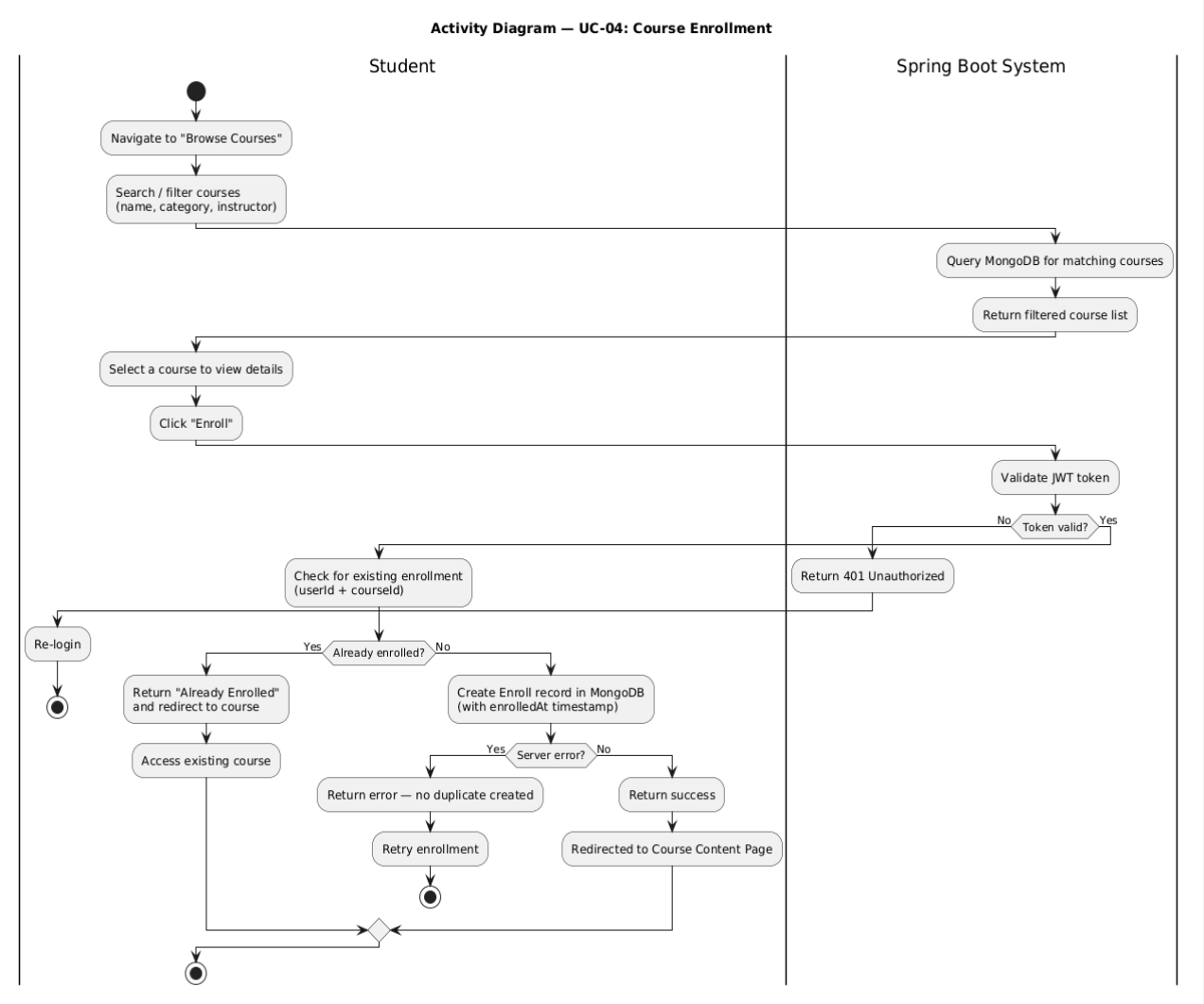


*Figure 4: Activity Diagram — UC-04: Course Enrollment*

## 5. UC-05 — Content Upload & Management

| Use Case Reference | |
|---|---|
| **Use Case ID** | UC-05 |
| **Use Case Name** | Content Upload & Management |
| **Primary Actor(s)** | Instructor |
| **Type** | Major |
| **Priority** | High |

**Description:** Covers the three content actions — Add Lecture, Edit Lecture, Delete Lecture — within a course the instructor owns. Swimlanes: Instructor | Spring Boot System.

**Notes:** *A parallel fork is used to represent the three possible content management actions available to the instructor.*

## [Activity Diagram — UC-05: Content Upload & Management]

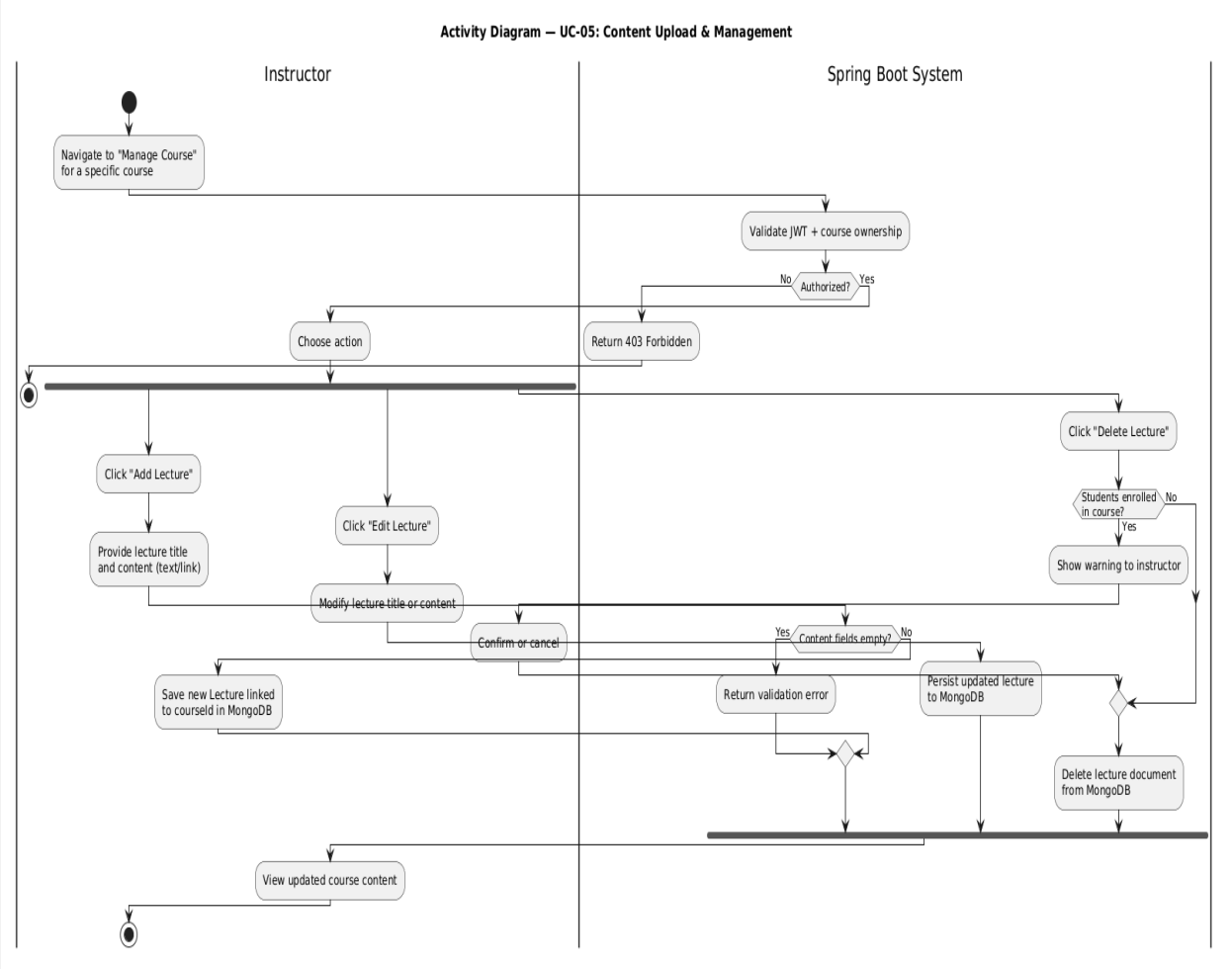


*Figure 5: Activity Diagram — UC-05: Content Upload & Management*

## 6. UC-06 — Learning Dashboard & Progress Tracking

| Use Case Reference | |
|---|---|
| **Use Case ID** | UC-06 |
| **Use Case Name** | Learning Dashboard & Progress Tracking |
| **Primary Actor(s)** | Student |
| **Type** | Minor |
| **Priority** | Medium |

**Description:** Shows dashboard loading, enrollment retrieval, progress calculation, lesson access, and progress upsert. Swimlanes: Student | Spring Boot System.

**Notes:** *Progress is stored as LessonProgress documents. Each lesson access triggers an upsert call.*

## [Activity Diagram — UC-06: Learning Dashboard & Progress Tracking]

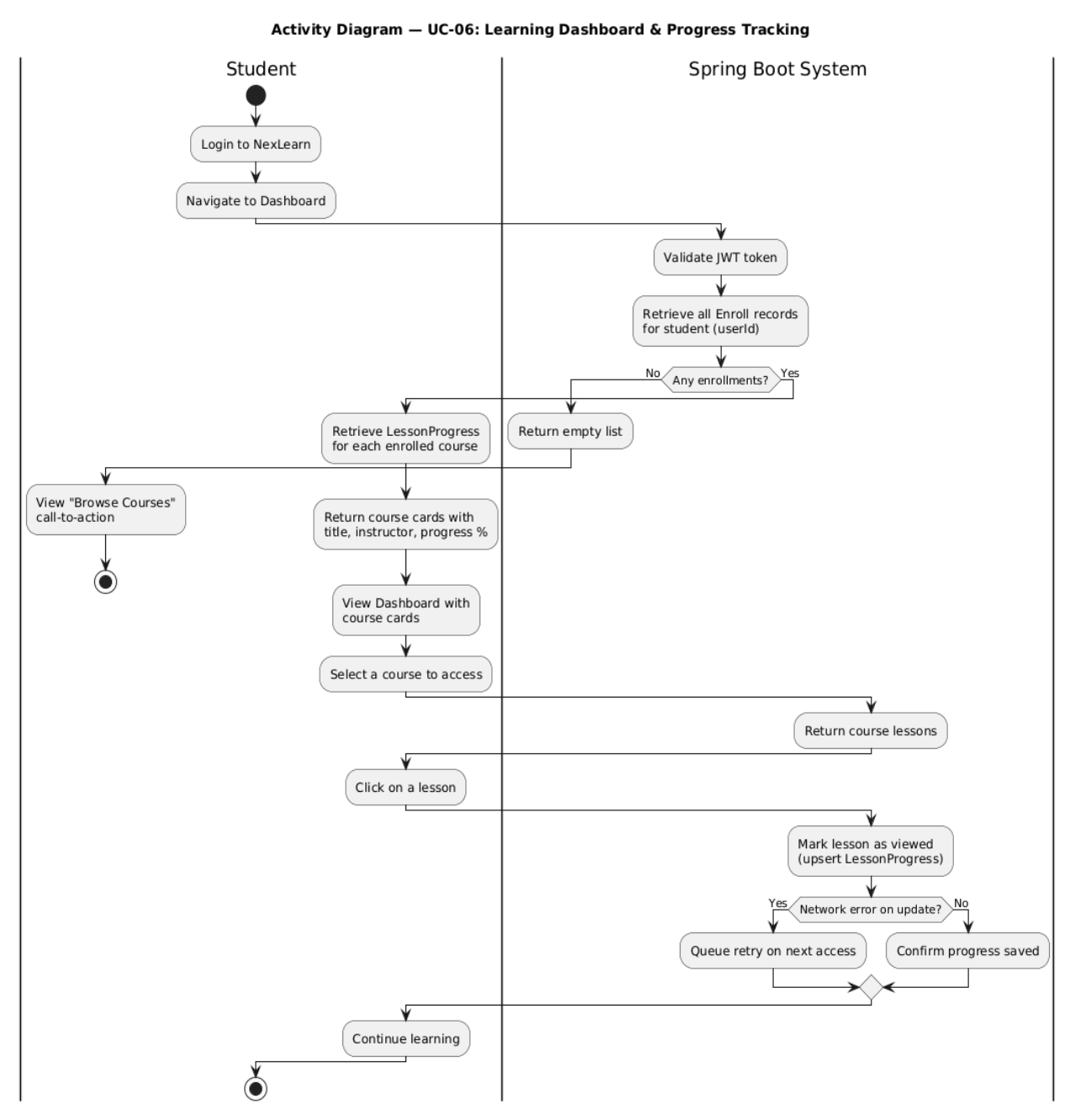


*Figure 6: Activity Diagram — UC-06: Learning Dashboard & Progress Tracking*

## 7. UC-07 — Student Monitoring by Instructor

| Use Case Reference | |
|---|---|
| **Use Case ID** | UC-07 |
| **Use Case Name** | Student Monitoring by Instructor |
| **Primary Actor(s)** | Instructor |
| **Type** | Minor |
| **Priority** | Medium |

**Description:** Depicts an instructor viewing enrolled students and their lesson-by-lesson progress for a specific course. Swimlanes: Instructor | Spring Boot System.

**Notes:** *The System verifies course ownership before returning any student data to protect privacy.*

## [Activity Diagram — UC-07: Student Monitoring by Instructor]

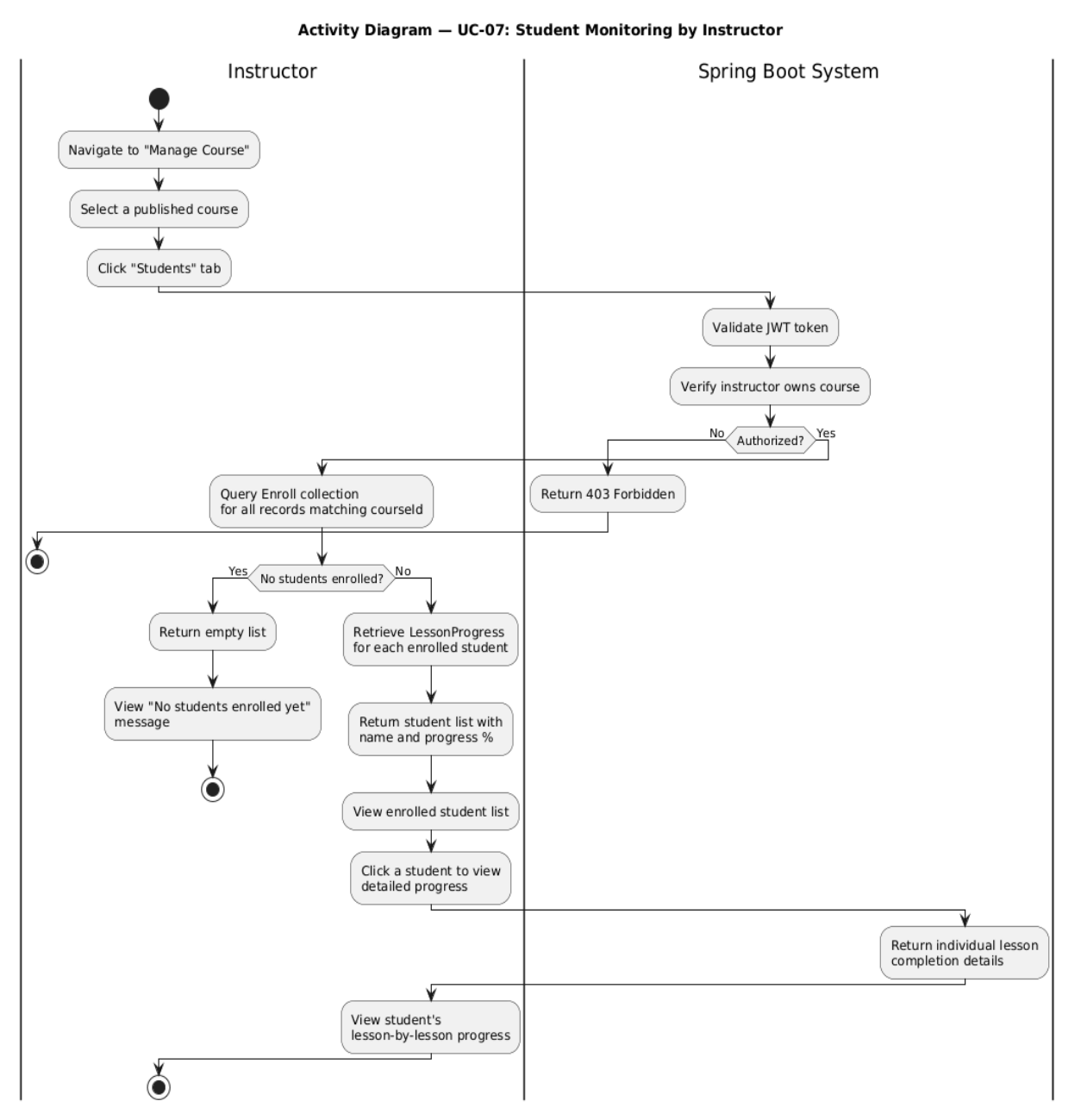


*Figure 7: Activity Diagram — UC-07: Student Monitoring by Instructor*

## 8. UC-08 — Search Courses

| Use Case Reference | |
|---|---|
| **Use Case ID** | UC-08 |
| **Use Case Name** | Search Courses |
| **Primary Actor(s)** | Student, Instructor |
| **Type** | Minor |
| **Priority** | Medium |

**Description:** Shows the search flow: empty query returns all courses; non-empty query triggers MongoDB regex search. Swimlanes: Student/Instructor | Spring Boot System.

**Notes:** *Search is performed on both title and description fields using a MongoDB regex query.*

[Activity Diagram — UC-08: Search Courses]

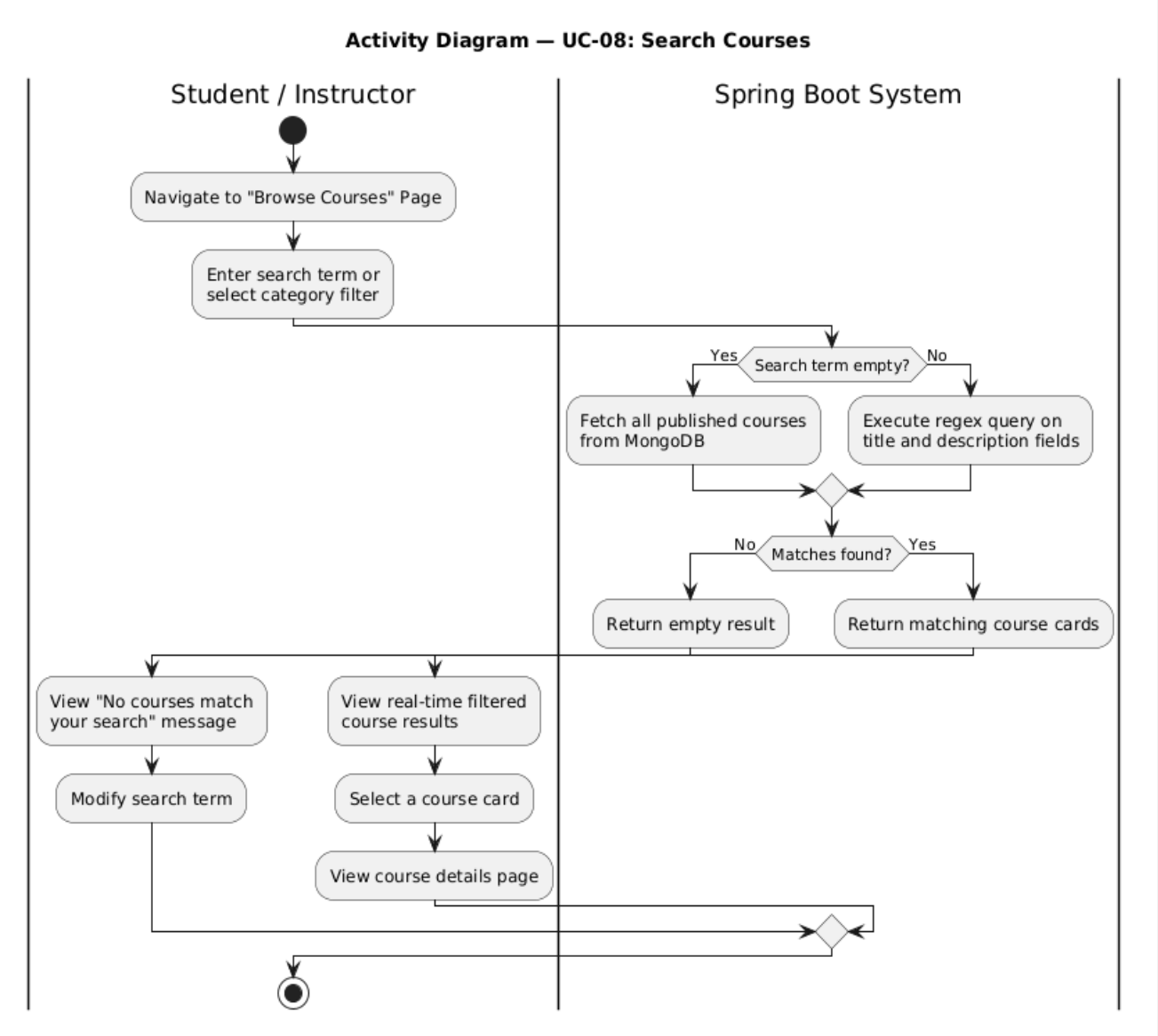


*Figure 8: Activity Diagram — UC-08: Search Courses*

## 9. UC-09 — User Profile Management

| Use Case Reference | |
|---|---|
| **Use Case ID** | UC-09 |
| **Use Case Name** | User Profile Management |
| **Primary Actor(s)** | Student, Instructor |
| **Type** | Minor |
| **Priority** | Low |

**Description:** Models profile fetch, field editing, password policy validation, email conflict detection, and MongoDB update. Swimlanes: Student/Instructor | Spring Boot System.

**Notes:** *Password is re-hashed on update. Email cannot be changed to one already in use by another account.*

## [Activity Diagram — UC-09: User Profile Management]

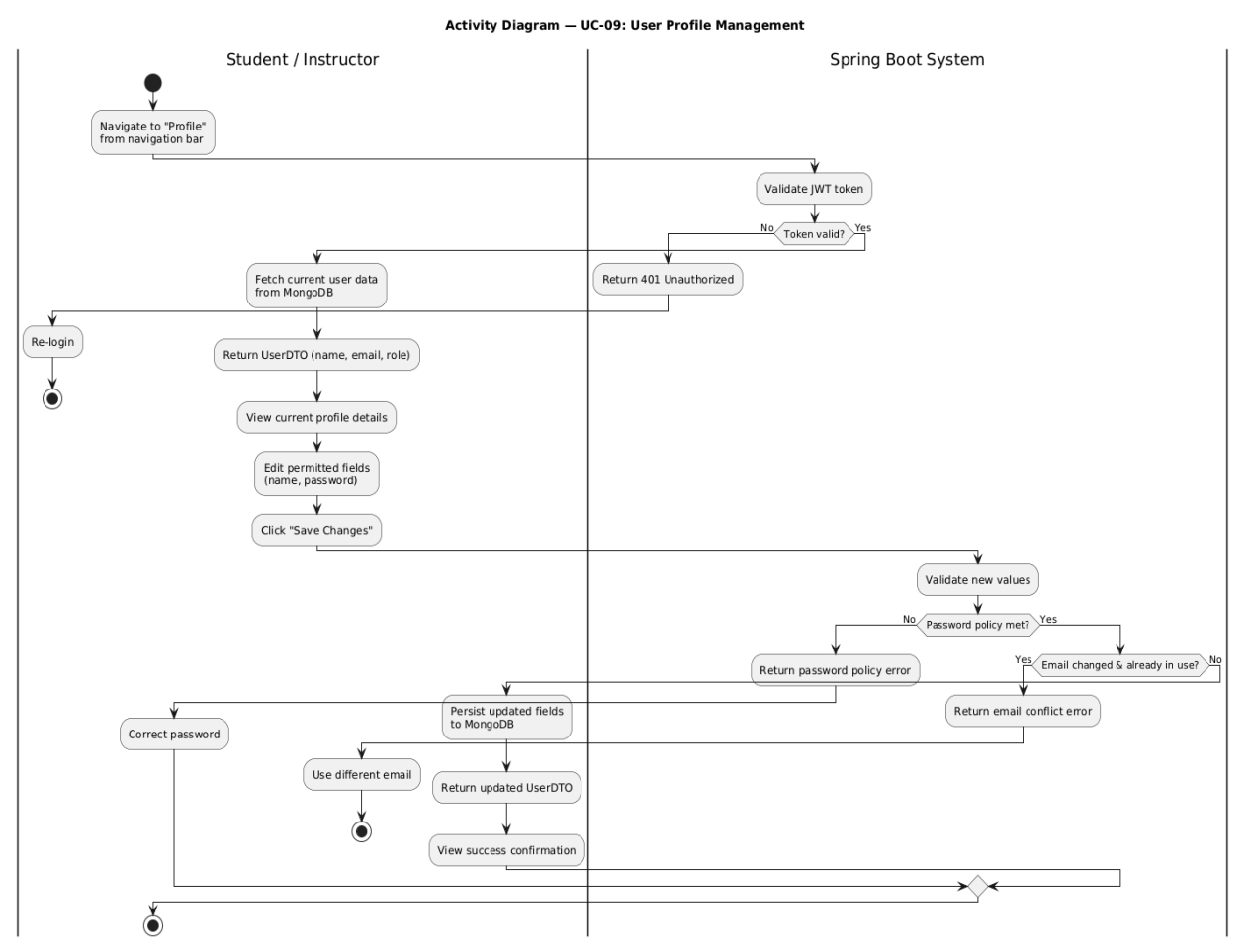


*Figure 9: Activity Diagram — UC-09: User Profile Management*

## 10. UC-10 — Contact & Support

| Use Case Reference | |
|---|---|
| **Use Case ID** | UC-10 |
| **Use Case Name** | Contact & Support |
| **Primary Actor(s)** | Student, Instructor, Anonymous |
| **Type** | Minor |
| **Priority** | Low |

**Description:** Shows form submission, field validation, and persistence of the ContactMessage to MongoDB. No authentication required. Swimlanes: User (Any/Anonymous) | Spring Boot System.

**Notes:** *This is the only use case accessible without authentication. Any visitor can submit a support message.*

[Activity Diagram — UC-10: Contact & Support]

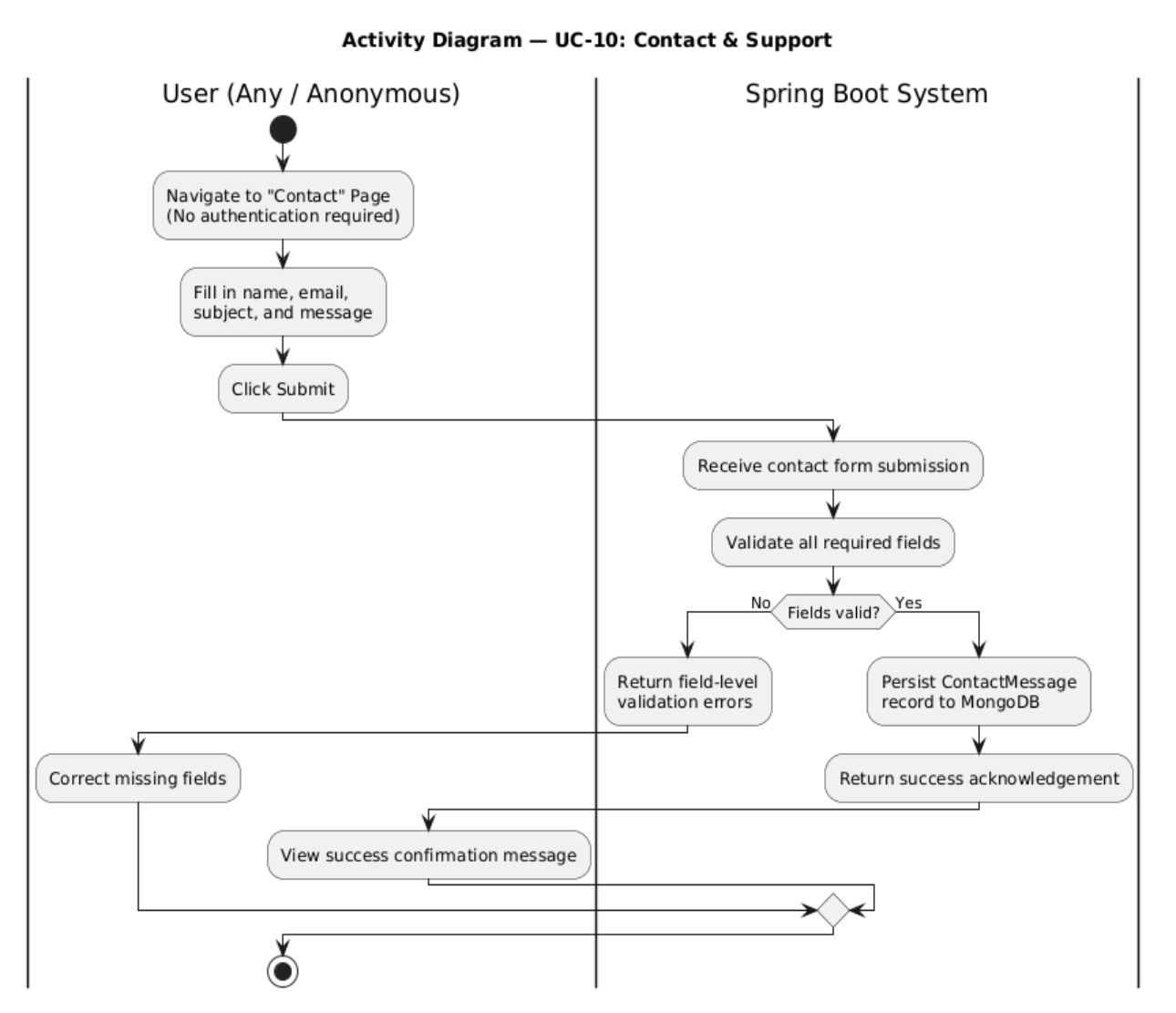


*Figure 10: Activity Diagram — UC-10: Contact & Support*